FactSense 3.0: System Architecture & Verification Pipeline

Complete Technical Documentation, Pipeline Workflow, Credibility Hierarchy & Evaluation

Course: NLP Academic Project	**Batch:** R7A	**Status:** Production Ready
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1. Executive Summary & Verification Philosophy

FactSense 3.0 is an automated evidence-grounded claim verification and fake news detection engine. Unlike conventional text-matching or sentiment-based fake news classifiers, FactSense is designed around the **Real-World Truth Principle**: the system strictly evaluates whether an assertion is factually authentic in the physical real world, never whether it is merely circulating or trending across social media.

Solving Circulation Confounding: When an input claims *'A Gazette circulating on social media, purportedly issued by the Ministry of Finance, declares every Saturday a holiday for PSU banks'*, typical search algorithms match thousands of viral social media posts confirming that the image is circulating. FactSense isolates the substantive authorization claim (*'Did the Ministry of Finance genuinely issue this order?'*) and consults primary regulatory portals (Tier 1) and certified fact-checkers (Tier 2). Because no official gazette was published, the system deterministically delivers a **REFUTED** or **MISLEADING** verdict.

Verdict	Color	Definition & Decision Threshold
SUPPORTED	Green	Corroborated by >= 1 Tier 1 (Official) or Tier 2/3 (Fact-Checker / Major News) source with no credible refutations.
REFUTED	Red	Directly denied or debunked by an official authority (Tier 1), certified fact-checker (Tier 2), or consensus credible reporting (Tier 3).
MISLEADING	Purple	Mixes true facts with false assertions, displays major headline-body framing exaggeration, or is rated 'Partly False' by fact-checkers.
INSUFFICIENT_EVIDENCE	Amber	The claim cannot be reliably confirmed or refuted across search engines (never defaulted to Supported).

2. Modular 8-Stage Pipeline Architecture

FactSense replaces static knowledge caches with eight decoupled, specialized processing stages:

- **Stage 1: claim_extractor.py:** Reads the entire input text. Performs headline detection, extracts up to 6 self-contained atomic claims, detects temporal scope (e.g., current, historical, 2026), and normalizes viral rumors into document authenticity questions.
- **Stage 2: query_generator.py:** Generates 4 diverse search query angles per claim: neutral, debunk ('fake', 'hoax', 'fact check'), official portal (using site: filters such as pib.gov.in, rbi.org.in), and certified fact-checkers (Alt News,

BOOM, Snopes).

- **Stage 3: web_search.py:** Concurrent live retrieval across Serper (Google), DuckDuckGo, and Wikipedia REST API. Features tiered Trafilatura page reading: parses full body text and publish date for top 3 hits, and falls back to search snippets for hits 4-8.
- **Stage 4: source_ranker.py:** Applies the 4-Tier Credibility Hierarchy. Computes dense semantic embeddings using all-MiniLM-L6-v2, strictly enforces a raw similarity threshold >= 0.35, applies a staleness penalty to outdated sources, and enforces Wikipedia background rules.
- **Stage 5: stance_classifier.py:** Batched LLM stance classification into 6 granular categories (affirms, denies, debunks_circulating_claim, partially_supports, mentions_without_verifying, neutral) with verbatim quote grounding. Lazy-loads DeBERTa-v3 NLI on demand.
- **Stage 6: framing_checker.py:** Evaluates headline vs. body text across a 9-point checklist (exaggeration, correlation as causation, unrepresentative sample, cherry-picked statistics, outdated info, misattributed photo/video) to detect contextual deception.
- **Stage 7: verdict_engine.py:** Executes deterministic decision rules, fact-checker override rules ('Partly False' -> MISLEADING), dynamic confidence calculation, and realistic 4-class probability distribution synthesis. Emits an early 'verdict' SSE event.
- **Stage 8: explainer.py:** Generates an 8-part grounded factual explanation with zero hallucinations: verdict summary, what is actually true (citing sources), why it is wrong/misleading, and unverified aspects.

3. Four-Tier Credibility Hierarchy

To ensure verifiable reliability, all retrieved passages are evaluated against strict domain criteria:

Tier	Domain & Suffix Patterns	Credibility Scope & Weight
Tier 1: Official Authorities	.gov, .gov.in, .nic.in, rbi.org.in, egazette.gov.in, whitehouse.gov, supremecourt.gov	**Highest Authority.** Sole authority on government orders, statutory gazettes, and official appointments. Overrides all other tiers.
Tier 2: Fact-Checkers	altnews.in, boomlive.in, snopes.com, factcheck.org, politifact.com, pib.gov.in/factcheck	**High Authority on Debunks.** Certified by the International Fact-Checking Network (IFCN). Overrides social media repetition.
Tier 3: News & Academic	reuters.com, apnews.com, bbc.com, thehindu.com, .edu, .ac.in, wikipedia.org, britannica.com	**Reputable Secondary Evidence.** Corroborates international events, science, and history. Wikipedia cannot sole-support breaking news.
Tier 4: General Web	reddit.com, twitter.com, x.com, instagram.com, medium.com, unverified commercial blogs	**Unverified Baseline.** Reflects viral circulation or public sentiment; cannot support a claim without corroboration from Tiers 1-3.

4. Deterministic Decision Synthesis & Overrides

Scenario / Evidence Profile	Verdict	Precedence Logic & Policy
>= 1 Tier 1/2 source affirms; 0 credible denials	**SUPPORTED**	Official record or certified investigation confirms authenticity.
>= 1 Tier 1 official denial exists	**REFUTED**	Official government or primary issuer denial overrides any number of Tier 4 claims.
>= 1 Tier 2 fact-checker debunk exists	**REFUTED**	Certified fact-checker debunks viral image, forged circular, or social media hoax.
Fact-checker rates 'Partly False' / 'Missing Context'	**MISLEADING**	Fact-checker override rule triggers Misleading unless a Tier 1 denial refutes it.
Headline framing severity is 'major'	**MISLEADING**	Headline asserts claims contradicted or unverified by the article's own body text.
Article contains both supported and refuted claims	**MISLEADING**	Mixed true/false content flagged; refuted parts listed in failed_parts.
Conflicting Tier 3 reporting (Affirm vs Deny)	**MISLEADING**	Major news agencies report contradictory information without consensus.
Only Wikipedia affirms news/document/statistic	**INSUFFICIENT**	Wikipedia background restriction; requires primary news or official records.
Only Tier 4 blogs or social media mention claim	**INSUFFICIENT**	Unverified claims without official or reputable news backing are never marked Supported.

5. Automated Acceptance Test Suite (tests/test_phase3.py)

FactSense is validated through 10 comprehensive acceptance test scenarios:

#	Test Scenario	Target Claim / Article Input	Expected Verdict	Status
1	Headline Exaggeration	AI tools make students score 40% higher (12-student survey)	**MISLEADING**	PASSED
2	Accurate Official Fact	Narendra Modi is the Prime Minister of India.	**SUPPORTED**	PASSED
3a	Viral Fake Gazette	Social media Gazette declares all Saturdays PSU holidays	**REFUTED**	PASSED
3b	Debunk Formulation	Same circular statement + 'is fake'	**REFUTED**	PASSED
4	False Implication	Ice cream sales cause shark attacks in July (correlation)	**MISLEADING**	PASSED
5	Misattributed Event	2026 video shows severe blizzard in downtown Cairo Egypt	**REFUTED**	PASSED
6	Fact-Checker Override	Fact-checker rates EV lifetime emissions 'Partly False'	**MISLEADING**	PASSED
7	Wikipedia News Rule	Breaking earthquake news found only on Wikipedia and blog	**INSUFFICIENT**	PASSED
8	Nonsense / Sci-Fi	Oceans of hot chocolate discovered on Jupiter moon	**INSUFFICIENT**	PASSED
9	No Headline Article	Multi-sentence paragraph without lead title	**SUPPORTED**	PASSED
10	Degraded Mode Cap	Mocked execution without external LLM API key	**degraded=True**	PASSED

6. Frontend Features & Performance Optimizations

• **Sub-8s Dual-Stage SSE Streaming:** An early verdict event is dispatched the moment Stage 7 completes. The browser immediately displays the verdict card, probability bars, and claim breakdown table while the slower grounded explanation (Stage 8) synthesizes in the background.
• **Headline Framing Card:** A side-by-side comparative card directly illustrates what the headline implies versus what the body text actually proves.
• **Degraded Mode Safety Banner:** If no LLM key is configured, FactSense warns users that it is running in local heuristic mode, capping maximum confidence at 60%.
• **Lazy-Loaded DeBERTa NLI:** Neural NLI models are lazy-loaded on demand (USE_DEBERTA_CROSSCHECK=false by default), ensuring instant server boot without memory lag.